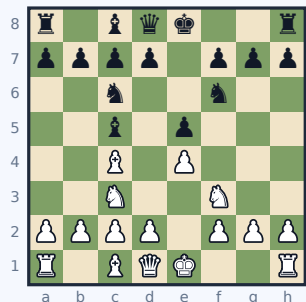


a Sensory encoding

FIXED

1

Board state S_t (FEN)

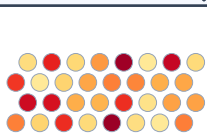


2



Piece-square
feature planes
 64×12

3



Input currents
 $1,546$ visual
neurons

b Connectome simulation

FROZEN WEIGHTS

4

$$\tau \dot{V}_i = -(V_i - V_{\text{rest}}) + \sum_j w_{ij} S_j(t)$$

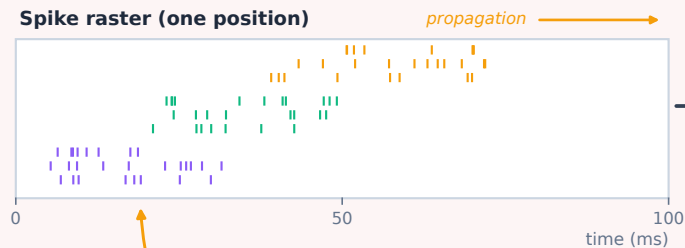
leaky integrate-and-fire; w_{ij} = FlyWire synapse counts (signed), no plasticity

Spike raster (one position)

Descending

Central

Visual



rates

output spikes

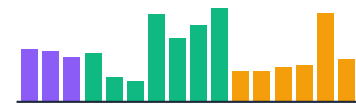
visual input

c Readout

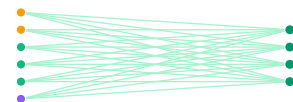
TRAINED

5

Firing-rate vector r



Linear readout



$$\hat{y} = Wr + b$$

Legal-move mask $\rightarrow$ argmax

Predicted
move



● Optic lobes / visual input

● Central brain

● Descending neurons

● Ventral nerve cord / motor

● Spiking neuron

Neurons drawn schematically